

Algebra II with Trigonometry

Chapter 11.4

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:

[4672988_alg-2trig-h-chp-11-4-note-docx-1.pdf](#)

1. **Shift Right or Left?** If x is replaced by $x - 4$ in an equation, how is the graph shifted?

2. **Shift Up or Down?** If y is replaced by $y + 3$, how is the graph shifted?

3. **Center of an Ellipse** What is the center of the ellipse

$$\frac{(x - 2)^2}{25} + \frac{(y + 1)^2}{9} = 1?$$

4. **Major Axis Direction** For the ellipse $\frac{(x + 3)^2}{4} + \frac{(y - 5)^2}{16} = 1$, is the major axis horizontal or vertical?

5. **Parabola Opening** Does $(x - 1)^2 = 8(y + 2)$ open up, down, left, or right?

6. **Parabola Vertex** What is the vertex of $(y - 4)^2 = -12(x + 2)$?

7. **Conic Type from Signs** For the equation

$9x^2 + 4y^2 - 18x + 8y - 11 = 0$, is the conic an ellipse, parabola, or hyperbola?

8. **Another Conic Type** For the equation $x^2 - 4y^2 + 2x - 8y + 1 = 0$, is the conic an ellipse, parabola, or hyperbola?

9. **Hyperbola Center** What is the center of $\frac{(x - 1)^2}{9} - \frac{(y + 2)^2}{16} = 1$?

10. **Focus and Directrix Idea** A parabola has focus $(1, 3)$ and directrix $x = 3$. Is the parabola opening left or right?